

Computational Physics

Summer 2026

Lecture #19

19. 6. 2026

Dylan Nelson

case 1: $\gamma=1$ "midpoint (Euler - Cauchy) method"

(3)

$$\Rightarrow a_1=0, a_2=1, b_{12}=\frac{1}{2}$$

$$k_1 = hf(t_i, y_i)$$

$$k_2 = hf(t_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1)$$

$$y_{i+1} = y_i + k_2 + \mathcal{O}(h^3)$$

case 2: $\gamma=1/2$ "modified Euler method"

$$\Rightarrow a_1=1/2, a_2=1/2, b_{12}=1$$

$$k_1 = hf(t_i, y_i)$$

$$k_2 = hf(t_i + h, y_i + k_1)$$

$$y_{i+1} = y_i + \frac{1}{2}(k_1 + k_2) + \mathcal{O}(h^3)$$

case 3: $\gamma=3/4$ "Heun's / Ralston's method"

case 4: $\gamma=0$ "Forward Euler"

- in general, RK2 schemes use two evaluations of $f(t, y)$ to advance a stepsize h , at $\epsilon_{\text{global}} \sim \mathcal{O}(h^2)$

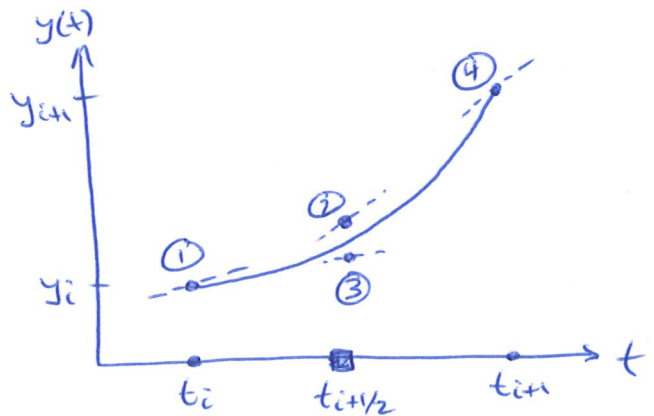
4th Order Runge-Kutta

$$k_1 = hf(t_i, y_i)$$

$$k_2 = hf(t_i + \frac{h}{2}, y_i + \frac{1}{2}k_1)$$

$$k_3 = hf(t_i + \frac{h}{2}, y_i + \frac{1}{2}k_2)$$

$$k_4 = hf(t_i + h, y_i + k_3)$$



$$y_{i+1} = y_i + \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4] + \mathcal{O}(h^5)$$

- uses four evaluations per step to achieve $\epsilon_{\text{global}} \sim \mathcal{O}(h^4)$

Generalized RK

④

- RK4 (with h) is more accurate than Forward Euler (with $h/4$)
- " " " " " RK2 (with $h/2$)
- [same number of function evaluations]

Thm:

Order	2	3	4	$5 \leq n \leq 7$	$8 \leq n \leq 9$	$10 \leq n$
best C	h^2	h^3	h^4	h^{n-1}	h^{n-2}	h^{n-3}

- general approach for RK method of any order:

(1) define n evaluation locations in h ($t_i \rightarrow t_{i+1}$)

$$t_j = t_i + h c_j, \quad j = 1, 2, \dots, n \quad c_j \in [0, 1]$$

(2) function evaluations start at t_i , so $c_1 = 0$

$$k_1 = f(t_i, y_i)$$

$$k_2 = f(t_i + c_2 h, y_i + h a_{21} k_1)$$

$$k_3 = f(t_i + c_3 h, y_i + h a_{31} k_1 + h a_{32} k_2)$$

$\vdots$

$$k_n = f(t_i + c_n h, y_i + h \sum_{j=1}^n a_{nj} k_j)$$

(3) update with weights b_j

$$y_{i+1} = y_i + h \sum_{j=1}^n b_j k_j$$

here, matrix a_{ij} is the RK matrix

b_i are the weights, $\sum b_i = 1$

c_i are the nodes

} information arranged in a "Butcher/RK tableau"

0					
c_2	a_{21}				
c_3	a_{31}	a_{32}			
$\vdots$	$\vdots$	$\vdots$	$\ddots$		
c_n	a_{n1}	a_{n2}	$\dots$	$a_{n,n-1}$	
	b_1	b_2	$\dots$	b_{n-1}	b_n

lower triangular with zero diagonal

$\Rightarrow$ explicit

otherwise,

$\Rightarrow$ implicit

Ex: RK4 tableau

0				
1/2	1/2			
1/2	0	1/2		
1	0	0	1	
	1/6	1/3	1/3	1/6

Forward Euler

0	
1	1

Modified Euler

0		
1	1	
	1/2	1/2

Midpoint

0	
1/2	1/2
	0
	1

Q: Backward Euler?

1	1
	1

non-vanishing diagonal $\Rightarrow$ implicit scheme

Adaptive Stepsize

- RK methods not time symmetric, need info only at t_i (beginning of step)
- $\Rightarrow$ can change h at each/any step, often required for physical problems

Ex:



interaction of a photon with H atom

$$\sigma(\nu) = \sigma_T \nu^4 [(\nu^2 - \nu_0^2)^2 + \gamma^2 \nu^2]^{-1}$$

has three regimes:

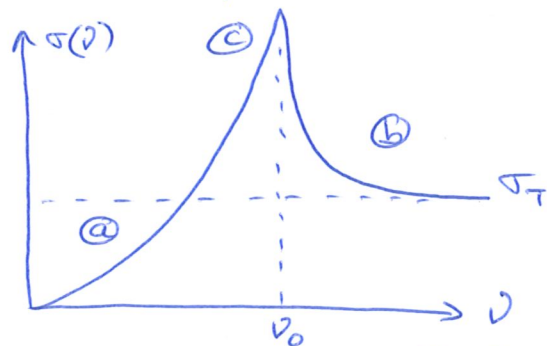
- Ⓐ low freq = Rayleigh scattering
- Ⓑ high freq = Thompson scattering
- Ⓒ resonance = steep Lorentz profile

$\hookrightarrow$ "stiff" equation $\equiv$ rapid changes, large derivatives

note: error terms always have higher derivatives of the solution

- if magnitude of derivative $\uparrow$ but solution does not, error can dominate

$\Rightarrow$ constant h is inefficient or inaccurate (or both)



Scattering cross section $[\text{cm}^2]$

[where $\sigma_T = \frac{8\pi}{3} r_0^2$ is the Thompson cross section
 $r_0 = e^2/mc^2$ is the electron radius
 $\gamma = 2e^2/3mc^2 \cdot \nu_0^2$ is the damping constant
 $\nu_0 = \Delta E/h$ is the resonant frequency]